MSIN0097 — UNIFIED AGENT SESSION LOG

Cardiovascular Risk Score Prediction | UCL MSc Business Analytics 2025–26 | Compiled: 27 Feb 2026 | Steps 1–6 Complete

SESSION METADATA	
Course / Module	MSIN0097 — Predictive Analytics | UCL MSc Business Analytics 2025–26
Dataset	cardiovascular_risk_dataset.csv — 5,500 patients, 15 features, target: heart_disease_risk_score (0–100)
Data Split	Train 70% (3,850) | Val 15% (825) | Test 15% (825) — stratified by Low / Medium / High risk bands
Session Dates	17 Feb 2026 (Steps 1–3) | 23 Feb 2026 (Step 3 corrections) | 27 Feb 2026 (Steps 4–6 + log)
Steps Covered	Steps 1–6: Problem Framing → EDA → Preprocessing → Model Exploration → Tuning → Final Presentation
Final Winner Model	MLP (Neural Network) — hidden_layer_sizes=(256,128,64), ReLU, Adam, early stopping (scaling fix applied)
Final Test MAE	~2.38 | Final Test R^2 ~0.98 | Macro F1 ~0.90 | High-risk Recall ~0.93
Total Errors Fixed	14 code/logic/presentation errors corrected across all steps and cells
Agent Mistakes	1 material: XGBoost declared winner without MLP in comparison (entry 5a) | 1 self-corrected: MLP scaling fix
Notebooks Produced	cardiovascular_eda.ipynb → _preprocessing.ipynb → _model_exploration.ipynb → _tuning_final.ipynb → _con
Log Format Note	All rows ordered by date + step. Merge future logs by inserting rows in date/step order. No agent column — un

SESSION ACTIVITY LOG

Each row = one task / prompt / fix. Ordered by date + step. Sources: agent_session_log_full.docx | agent_session_log_step5.docx | agent_session_log.docx — deduplicated and merged.

DATE	LEVEL	STAGE	MESSAGE	OUTPUT / DETAILS
► STEP 1 — PROBLEM FRAMING | 17 Feb 2026				
17-02-26	COMPLETED	**Step 1 — Problem Framing**	Frame the ML problem: classification vs regression, target variable, metrics, constraints, fairness	Recommended regression on heart_disease_risk_sco hierarchy: MAE primary → RMSE/R^2 secondary → M risk_category leakage risk.
17-02-26	EXPLAINED	**Step 1 — Concepts**	Explain Macro F1 and its working	Explained Precision, Recall, F1 from ground up. Show Numerical example: Low=0.89, Med=0.80, High=0.59
17-02-26	EXPLAINED	**Step 1 — Concepts**	Why avoid plain accuracy?	Class imbalance: model predicting all Medium gets ~4 performance. Macro F1 penalises that equally.
17-02-26	COMPLETED	**Step 1 — Target Confirmation**	Confirmed risk_category is algorithmic bucketing of heart_disease_risk_score — regression finalised	Framing finalised. risk_category excluded entirely (lea score. Derived classification (Low/Med/High) for repor
► STEP 2 — EDA | 17-23 Feb 2026				
17-02-26	COMPLETED	**Step 2 — EDA Build**	Build full EDA notebook (10 sections): data quality, target distribution, feature distributions, outliers, categorical, correlations, scatter, VIF, boundary analysis	Delivered cardiovascular_eda.ipynb — 34 cells, 10 se target, systolic_bp r=0.90, alcohol outliers 3.9%, ~25%
23-02-26	FIXED	**Step 2 — EDA Fix**	VIF multicollinearity interpretation wrong — age_bmi should not appear, leads to false interpretation	Corrected: only systolic_bp (VIF=11.2) crosses thresh moderate physiological correlations, not errors. daily_ clean. age_bmi excluded from linear pipeline VIF by c
23-02-26	COMPLETED	**Step 2 — EDA Fix**	Generate corrected VIF code chunk only — do not regenerate full notebook	Delivered corrected VIF cell code only. Scoped fix to c unnecessary file rewrite.
23-02-26	EXPLAINED	**Step 2 — Concepts**	Explain significance of boundary region analysis for later project stages	4-stage utility: (1) Error stratification in evaluation — b failure. (2) Separate MAE reporting core vs boundary. focus — boundary patients most informative for attribu
► STEP 3 — PREPROCESSING | 19-23 Feb 2026				
19-02-26	COMPLETED	**Step 3 — Preprocessing Build (3a)**	Deliberately minimal prompt: asked for full preprocessing pipeline and data split with no ratios or specifics given	Accepted. 70/15/15 stratified split, custom Winsoriser sklearn pipelines (linear + tree), leakage guard cell, W check. Split ratios, fit-on-train-only discipline, and veri explicitly compares scaler means to training vs full da
19-02-26	UPDATED	**Step 3 — Encoding (3b)**	Categorical encoding applied to full df_encoded before train/val/test split — structural concern	Modified. While ordinal/binary encoding is determinist practice places encoding inside the sklearn pipeline p structural concern — acceptable for this project but no
19-02-26	CORRECTED	**Step 3 — Feature Engineering (3c)**	No feature engineering added — both pipelines pass raw features with only scaling/Winsorisation differences	Modified. Dataset has zero preprocessing complexity Verified age × bmi interaction (r=0.838) outperforms e r=0.71). Added as engineered feature. After VIF analy restricted age_bmi to tree pipeline only.
19-02-26	CORRECTED	**Step 3 — CV Setup (3d)**	No cross-validation setup — only holdout validation set provided	Rejected. Single holdout val set insufficient for robust rows. Added StratifiedKFold(n_splits=5) on training da reserved for final evaluation only.

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19-02-26	FIXED	**Step 3 — StratifiedKFold Bug (3e)**	Executed StratifiedKFold cell — agent referenced X_train_tree inside CV fold verification loop — NameError on execution	Bug caught during execution by reading code carefull cells — variable does not exist at that point. Correcte indices, not transformed arrays. Demonstrates agent notebook cells.
23-02-26	COMPLETED	**Step 3 — Preprocessing Reconstruction**	Rebuild complete preprocessing pipeline with all 5 corrections incorporated into final notebook	Delivered Preprocessing_fixed_(2).ipynb — 59 cells. (13 features). Pipeline B: passthrough (15 features + verified. All 16 readiness checks passed. StratifiedKF
► STEP 4 — MODEL EXPLORATION | 27 Feb 2026				
27-02-26	COMPLETED	**Step 4 — Model Exploration Build**	Build 4 models: (1) Linear Regression baseline on linear arrays, (2) RF modified hyperparams, (3) XGBoost primary candidate, (4) MLP 3 hidden layers with scaling fix	Added 28 cells (59–86). All 4 models with residuals, C RF, loss curve for MLP. Comparison table + 3-panel b importance + rank divergence table. 5 accepted / 2 m register.
27-02-26	CORRECTED	**Step 4 — MLP Scaling Fix**	Agent used X_train_tree (unscaled) for MLP despite its own markdown flagging scale-sensitivity	StandardScaler fitted on X_train_tree only (no leakag X_val_tree_scaled / X_test_tree_scaled. All MLP arra X_train column means. Noted as self-correction in de
27-02-26	EXPLAINED	**Step 4 — Concepts**	Explain MLP functionality: architecture, backpropagation, why it typically loses to XGBoost on tabular data	Explained: single neuron mechanics (weighted sum + optimiser + backprop, 3 regularisation mechanisms (L vs XGBoost on tabular: no inductive bias, scale sensi
► STEP 5 — TUNING (BUILD PASS) | 27 Feb 2026				
27-02-26	COMPLETED	**Step 5 — Tuning Build**	Tune XGBoost (60 iter, 9 params) and RF (40 iter). Winner by CV MAE. Deep error analysis + test evaluation + end-to-end comparison	Added 38 cells (91–128) to model_exploration.ipynb. winner_model auto-selected by CV MAE. Error analys boundary vs core, feature quartile heatmap, worst-20 (raw + row-normalised). End-to-end comparison table
► STEP 5 — REVIEW & AUDIT PASS | 27 Feb 2026 (cardiovascular_tuning_final.ipynb)				
27-02-26	FIXED	**Step 5 — Tuning 1.2 (Cell 95)**	groups=y_train_bands passed to xgb_search.fit() — TypeError with StratifiedKFold	groups= belongs to GroupKFold, not StratifiedKFold. object. Fix: remove groups=y_train_bands from .fit() e
27-02-26	FIXED	**Step 5 — Tuning 1.3 (Cell 98)**	Same groups= bug copy-pasted to rf_search.fit()	Same fix applied: groups=y_train_bands removed fro tuning cells now clean.
27-02-26	ADDED	**Step 5 — Error Analysis (Cell 106)**	Abs error vs predicted scatter absent — brief explicitly requires it; residuals-only masks heteroscedasticity	4th panel added: abs_error vs predicted with rolling m Heteroscedasticity check prints mean abs error for lov errors cannot cancel in abs error view.
27-02-26	ADDED	**Step 5 — Three-Way Comparison (Cell 125 new)**	Three-way comparison (Val/CV/Test) not assembled — numbers scattered, never unified	New Cell 125: prints MAE, RMSE, R² across Val Set | columns. Gap analysis flags CV vs Val, Val vs Test, C guide included.
27-02-26	FIXED	**Step 5 — End-to-End Table (Cell 126)**	Test MAE stored under Val MAE column — agent acknowledged in comment but left mislabelled	Cell 126 rebuilt: dedicated Test MAE column added. N Three-panel chart: Val MAE | CV MAE with error bars unevaluated). Winner highlighted green.
27-02-26	FIXED	**Step 5 — Test Eval (Cell 124)**	Boundary margins hardcoded as 20, 30, 50, 60 instead of MARGIN variable (defined as 5)	Replaced hardcoded values with 25-MARGIN, 25+MA Consistent with val analysis and resilient to MARGIN
27-02-26	BUILT	**Step 5 — Full Reconstruction**	Reconstruct complete notebook Steps 1–5 with all corrections and missing features applied	Delivered cardiovascular_complete_corrected.ipynb – corrections applied (groups= ×2, abs error panel, thre boundary MARGIN fix). MLP scaling fix from Step 4 c
► STEP 5 — CORRECTION PASS | 21 Feb 2026 (cardiovascular_complete_corrected.ipynb Cells 82–113)				
21-02-26	FIXED	**Step 5 — Tuning 1.2 (Cell 82)**	XGBoost RandomizedSearchCV failing — StratifiedKFold incompatible with continuous regression target	Fix: replaced cv=cv (StratifiedKFold) with cv=kf (KFol KFold(n_splits=5, shuffle=True, random_state=42).
21-02-26	FIXED	**Step 5 — Tuning 1.3 (Cell 85)**	RF RandomizedSearchCV same error as XGBoost	Fix: cv=kf applied to Cell 85 (RF search). Same KFold needed.
21-02-26	CORRECTED	**Step 5 — Tuning 1.4 (Cell 88)**	Winner selection Cell 88 only compared XGBoost Tuned vs RF Tuned — MLP excluded entirely from competition	Material mistake (logged as entry 5a). Cell 88 rewritte (Neural Network)", winner_model=mlp, winner_preds_ winner_X_test=X_test_tree_scaled. MLP Val MAE 2.3 gap on every metric.
21-02-26	FIXED	**Step 5 — Test Eval (Cell 107)**	f1_score crash: pd.Categorical dtype incompatible with sklearn classification metrics	Fix: .astype(str) added on both y_test_bands and y_p Categorical → plain strings that sklearn accepts.
21-02-26	FIXED	**Step 5 — Three-Way (Cell 110)**	tuning_summary.loc[winner_name] KeyError — MLP not in tuning_summary (only XGBoost Tuned and RF Tuned)	Fix: all winner metric lookups changed from tuning_su results_df.loc[winner_name] in Cells 110 and 122. ML
21-02-26	FIXED	**Step 5 — End-to-End Table (Cell 112)**	Test MAE being stuffed into Val MAE column	Cell 112 rewritten: separate Test MAE column added. behaviour — test set touched once for winner only).
21-02-26	FIXED	**Step 5 — Chart (Cell 113)**	Third chart (Test MAE) had compressed y-axis vs first two charts — misleading visual	Fix: axes[2].set_ylim(axes[0].get_ylim()) added after y y-axis range.
21-02-26	FIXED	**Step 5 — Chart (Cell 74)**	Orange bar was on XGBoost (3rd bar) — should be on MLP (4th bar, winner)	Fix: bar_colors = [PALETTE, "#5BA4C7", "#8EC3DC" index 3 = MLP. Matches the legend description.

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21-02-26	FIXED	**Step 5 — Cell 86 Exec Order**	res_rf_tuned NameError in Cell 88 despite Cell 86 RF search completing successfully	Root cause: extract+evaluate cell between RF search Provided full cell: rf_tuned = rf_search.best_estimator then Cell 88.
► STEP 6 — FINAL PRESENTATION | 27 Feb 2026				
27-02-26	FIXED	**Step 6 — Cell C (Decision Table)**	KeyError "MLP Neural Network" — lines 30–32 still had old key string without parentheses	Entire Cell C regenerated. All 4 .loc[] references now with parentheses. Also updated rationale printout to e
27-02-26	UPDATED	**Step 6 — Cell 116 (Markdown)**	Markdown still declared XGBoost (Tuned) as winner with XGBoost-based rationale	Rewritten: MLP declared winner. Criteria: Val MAE → all four. Agent mistake (log 5a) explicitly acknowledge
27-02-26	UPDATED	**Step 6 — Cell 121 (Model Card Print)**	Model card hardcoded XGBoost architecture, framework, feature importances throughout	Rewritten: model type = MLP, architecture = Dense 2 framework = scikit-learn MLPRegressor scaled pipelin epoch 39) and interpretability caveat (no intrinsic impo
27-02-26	FIXED	**Step 6 — Cell 122 (Model Card Visual)**	feature_importances_ crash — MLP has no intrinsic importances. Two bottom panels overlapping.	Rewritten: GridSpec 3×3 (18×14). feature_importance (n_repeats=10, val set). MLP loss curve added as Pa importance takes full bottom row (row 3 all cols). No c
27-02-26	COMPLETED	**Step 6 — Cell 118 (Limitations Markdown)**	Convert code-based limitations/next steps explanation to clean markdown cell	Markdown generated: L1–L7 limitations (added L7 ML importances). 3-tier next steps: HIGH (real EHR valid MEDIUM (boundary-aware loss, temporal split) | LOW
► AGENT DECISION LOG + FINAL AUDIT | 27 Feb 2026				
27-02-26	LOGGED	**Agent Decision Log (Entry 5a)**	Log material mistake: XGBoost declared winner without including MLP in winner selection comparison	New row 5a added to Assessment_Information_Cours MAE 2.3998 vs XGBoost Tuned 2.6629 (10% gap). F before test evaluation. All Step 5 and Step 6 cells re-r
27-02-26	CLEAN	**Final Notebook Audit**	Full 122-cell notebook audit: errors, redundant code, misleading comments, inefficiencies	4 issues found (no runtime errors): (1) 3 vars duplicat ORDINAL_FEATURES, BINARY_FEATURES alread StandardScaler as TreeScaler in Cell 69 (already imp 80 (says StratifiedKFold but tuning now uses KFold). StratifiedKFold). All flagged for correction.
27-02-26	COMPLETED	**Session Log Compilation**	Compile all 3 agent session logs into one unified chronological document	Merged agent_session_log.docx + agent_session_log agent_session_log_full.docx. Deduplicated overlappin activity rows across Steps 1–6 (17 Feb – 27 Feb 2026 included.